# LINK: Learning to Localize from Known to Unknown Scenes

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## Abstract

LiDAR relocalization based on scene coordinate regression (SCR) degrades sharply when test trajectories traverse route segments absent from training, a practical challenge in long-range autonomous-driving deployment, where complete route coverage is rarely achievable. We present LINK, a confidence-guided framework for relocalization under partial scene coverage. LINK predicts an absolute pose hypothesis and scene features with an SCR backbone, estimates pose reliability from temporal pose–feature sequences, and uses a decision policy to arbitrate between direct absolute hypotheses and relative motion propagated from reliable historical estimates, thereby preserving globally referenced localization across uncovered segments. We also introduce UrbanUnseen, a city-scale benchmark with native known-to-unknown scene transitions, together with controlled partial-coverage protocols on Oxford and NCLT. Experiments show that LINK substantially improves robustness when training coverage is incomplete, consistently outperforms representative map-free baselines under partial coverage, and remains effective in fully known scenes.

## 1 Introduction

Reliable LiDAR relocalization is a fundamental capability for robots and autonomous vehicles operating in large-scale outdoor environments [44, 46, 50]. Existing approaches are commonly divided into map-based and map-free paradigms [17, 40, 43, 23, 41]. Map-based methods localize by matching the current scan against a pre-built 3-D map [29, 48], but storing, maintaining, and querying explicit maps and features introduces nontrivial deployment costs in long-range applications [8]. Map-free methods avoid explicit map access at test time and instead infer globally referenced pose hypotheses directly from LiDAR observations [47]. Among them, scene coordinate regression (SCR) pipelines [12, 32, 21, 23, 41] have shown promising accuracy and efficiency for outdoor relocalization.

However, existing map-free SCR methods still rely on an implicit assumption: the training trajectories sufficiently cover the scenes encountered during inference [15, 30, 24, 52]. Exhaustive route coverage is expensive and often impractical, so a test trajectory may traverse road segments absent from training [3, 7]. We refer to this setting as partial scene coverage, and denote training-covered and training-uncovered route segments as known and unknown scenes, respectively. Under known-to-unknown transitions, direct SCR may still output an absolute pose, but the prediction can become unreliable because the model must extrapolate beyond the scene distributions it has learned [4, 15, 30, 51]. The central challenge is therefore not merely improving regression accuracy in known scenes, but maintaining globally referenced localization when coverage changes along a route. More generally, scene membership is only a proxy for pose quality: SCR hypotheses may also become unstable on nominally covered long routes. Fig. 1 illustrates the issue: under the partially known NCLT protocol, LightLoc [23] drifts severely after entering training-uncovered segments.

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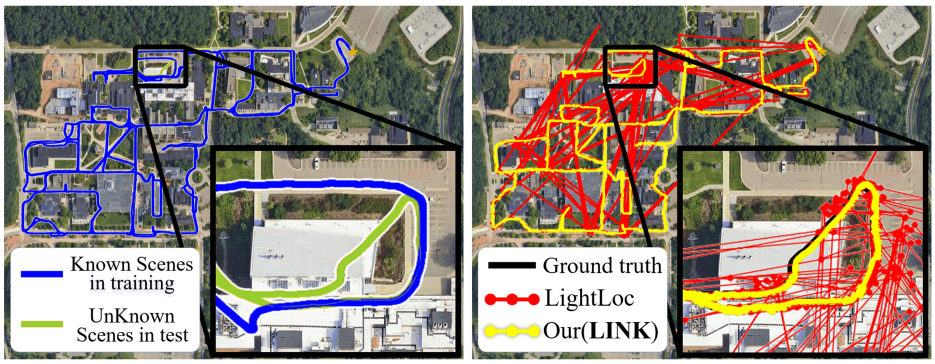



Figure 1: The impact of unknown scenes on LiDAR localization. Under a partially known NCLT protocol, some test segments are absent from training. LightLoc [23] produces frequent errors in unknown regions, whereas LINK maintains better trajectory consistency.

Addressing this challenge requires two coupled capabilities. First, the system must estimate whether the current absolute pose hypothesis remains trustworthy. Unreliable SCR hypotheses are difficult to identify from a single scan alone, since failures often appear as temporal instability rather than obvious per-frame appearance changes; temporal cues have therefore become important in localization [14, 53, 45, 13]. Second, once confidence degrades, the system should decide whether to accept direct SCR localization or propagate from a reliable historical pose. This is a sequential problem: switching depends on confidence history, reference-pose availability, and accumulated odometric drift. A practical solution should therefore combine temporal pose-reliability estimation with sequential mode selection.

To this end, we present LINK, a confidence-guided hybrid relocalization framework for map-free LiDAR relocalization under partial scene coverage. LINK uses an SCR backbone to produce absolute pose hypotheses and scene features, estimates pose reliability from temporal pose–feature consistency as a soft score rather than a hard known/unknown label, and uses a decision policy to choose between direct SCR localization and odometric propagation from reliable historical poses. In this way, LINK maintains a globally referenced trajectory across uncovered segments while suppressing unstable SCR hypotheses.

Existing evaluation protocols are also insufficient for assessing robustness under partial scene coverage. Public outdoor localization benchmarks [3, 7, 6, 16] do not explicitly separate training-covered and training-uncovered route segments. We therefore introduce UrbanUnseen, a city-scale LiDAR relocalization benchmark with route-level known/unknown transitions, and establish known/unknown protocols on Oxford [3] and NCLT [7] under random and manual partitions. Throughout the paper, Oxford denotes the Oxford Radar RobotCar benchmark with trajectory-aligned ground-truth poses following SGLoc [21]; Appendix A.1 provides the details. Together, these settings cover fully known routes, controlled partial coverage, and route-level known/unknown transitions.

The main contributions of this work are summarized as follows:

- We formulate map-free LiDAR relocalization under partial scene coverage and establish known/unknown evaluation protocols on Oxford and NCLT, including both random and manual partitions.
- We propose LINK, a confidence-guided hybrid relocalization framework that estimates pose reliability from temporal pose–feature consistency and adaptively switches between direct SCR localization and odometric propagation from reliable historical poses.
- We introduce UrbanUnseen and provide systematic evaluations on Oxford, NCLT, and UrbanUnseen, showing that LINK improves robustness under incomplete coverage while remaining competitive in fully known scenes.

## 2 Related Work

**Relocalization paradigms.** LiDAR relocalization methods can be broadly grouped into map-based and map-free paradigms. Map-based methods estimate the global pose by matching a query scan

against a pre-built representation, typically retrieving candidate places and then refining the pose by registration. PointNetVLAD [33], MinkLoc3D [20], SOE-Net [38], and RING++ [40] learn or design global descriptors for place retrieval, while SpinNet [1], TEASER++ [42], and Buffer [2] improve geometric verification and registration. These methods can be accurate, but explicit map storage, maintenance, and search introduce deployment costs in long-range settings [8, 43]. Map-free relocalization avoids explicit map access by directly predicting globally referenced poses from LiDAR observations. APR methods such as PointLoc [36], STCLoc [45], HypLiLoc [35], NIDALoc [47], and DiffLoc [22] regress poses directly, while SCR methods build on scene-coordinate prediction [32, 12, 5]. Recent outdoor SCR systems, including SGLoc [21], LiSA [41], and LightLoc [23], improve accuracy and efficiency without explicit maps. However, these map-free methods are still mostly evaluated under the assumption that training and test routes share sufficient scene coverage, leaving their behavior under incomplete coverage less understood.

**Reliability and hybrid localization.** Prior work has long recognized that pose predictions need reliability estimates. Kendall and Cipolla [14] modeled predictive uncertainty for pose regression, KFNet [53] incorporated temporal filtering into relocalization, and STCLoc [45] showed that temporal consistency can improve LiDAR localization. Reliability cues also arise in retrieval and registration: Vidanapathirana et al. [34] re-ranked retrieved scans through spectral geometric verification, while Huang et al. [13] used consistency-aware modeling to improve registration reliability. These studies suggest that temporal instability and geometric inconsistency are informative cues for deciding whether a current pose estimate should be trusted. Practical systems also often combine absolute and relative localization. Voxelized GICP [19] provides efficient relative motion estimation, and graph-based optimization [37] improves long-horizon trajectory stability. However, such hybridization usually assumes either explicit maps or sufficiently reliable absolute updates. Under partial scene coverage, the key question becomes when a map-free absolute hypothesis should be accepted and when localization should instead be propagated from a reliable historical pose. This decision is not well captured by frame-wise confidence thresholding because its consequence depends on confidence history, reference-pose availability, and accumulated odometric drift. Existing work therefore provides useful ingredients for reliability estimation and trajectory stabilization, but reliability-guided absolute/relative mode selection as a sequential decision problem for map-free LiDAR relocalization remains less directly explored.

**Benchmarks and protocols.** Oxford [3, 21] and NCLT [7] remain the standard repeated-route benchmarks for outdoor map-free LiDAR relocalization. SGLoc [21], LiSA [41], DiffLoc [22], and LightLoc [23] all report their primary results on these datasets. Boreas [6] expands seasonal and spatial diversity, whereas Mulran [16] supports urban localization and place-recognition research. However, these benchmarks do not explicitly separate training-covered and training-uncovered route segments. As a result, robustness to partial scene coverage is usually tested only indirectly, for example by repartitioning existing trajectories. This distinction matters because a method may remain competitive in fully known scenes yet fail once the route enters unknown segments. A complementary evaluation setting is therefore needed to cover conventional fully known relocalization, controlled partial-coverage protocols, and native route-level known-to-unknown transitions.

# 3 Method

## 3.1 Overall Framework

Under partial scene coverage, a test trajectory may traverse both training-covered and training-uncovered route segments. Our objective is to output a globally referenced pose for every frame while reducing large localization errors once direct SCR localization becomes unreliable. The overall pipeline is shown in Fig. 2.

Given the current LiDAR scan $\mathbf{x}_t$, the SCR backbone predicts an absolute pose hypothesis $\mathbf{p}_t^{\mathrm{scr}} = (\mathbf{t}_t^{\mathrm{scr}}, \mathbf{q}_t^{\mathrm{scr}})$ and an intermediate feature map $\mathbf{F}_t \in \mathbb{R}^{N\times C}$, where $N$ and $C$ denote the number of feature tokens and the feature-channel dimension. We instantiate the backbone with LightLoc [23]; training details are provided in Section 5.1. The predicted pose provides an efficient map-free absolute hypothesis, while the feature map is retained for temporal reliability estimation.

A reliability estimator produces a soft score $c_t \in [0, 1]$ from a temporal pose–feature sequence. The decision policy then selects one of two localization modes: accepting the current SCR hypothesis, or

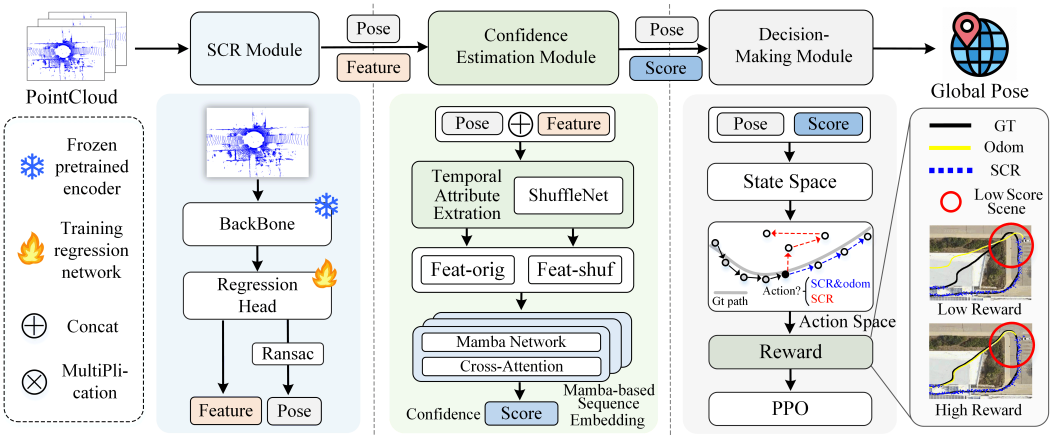



Figure 2: Overview of LINK. For each incoming LiDAR scan, the SCR backbone predicts an absolute pose hypothesis together with scene features. A temporal reliability estimator evaluates whether that hypothesis remains trustworthy, and the decision policy then chooses between direct SCR localization and odometric propagation from a reliable historical pose. The reward and PPO branches are used only during policy training.

propagating a stored reliable pose through relative odometry. The final output pose is:

$$\hat{\mathbf{p}}_t = \begin{cases} \mathbf{p}_t^{\mathrm{scr}}, & a_t = 0, \\ \mathbf{p}_{\tau_k}^{\mathrm{ref}} \oplus \Delta\mathbf{p}_{\tau_k \to t}^{\mathrm{odom}}, & a_t = k,\ k \in \{1, \ldots, K\}, \end{cases} \tag{1}$$

where $a_t$ is the action selected by the policy, $K$ is the number of candidate reference poses, $\mathbf{p}_{\tau_k}^{\mathrm{ref}}$ is the $k$-th candidate reference pose stored in a fixed-size buffer of recent high-confidence poses, $\Delta\mathbf{p}_{\tau_k \to t}^{\mathrm{odom}}$ is the odometric motion from time $\tau_k$ to the current frame $t$, and $\oplus$ denotes rigid-body pose composition. Thus, LINK remains map-free at inference while reducing exclusive reliance on SCR predictions in uncovered route segments.

### 3.2 Temporal Pose-Reliability Estimation

Unreliable SCR predictions under partial scene coverage are often exposed by temporal instability: pose hypotheses may fluctuate or drift even when individual scans remain locally plausible. We therefore estimate reliability from the ordered evolution of pose hypotheses and scene descriptors instead of from a single scan. Because this reliability score is evaluated online for every frame, the temporal module should capture pose–feature dynamics with limited inference overhead. We use shared Mamba blocks to model the ordered pose–feature sequence, followed by cross-attention fusion to compare the original stream with a lightweight perturbed companion stream.

For each frame, the intermediate feature map is compressed by a learnable operator $G(\cdot)$ and concatenated with the normalized SCR pose:

$$\begin{aligned} \mathbf{f}_t &= G(\mathbf{F}_t) \in \mathbb{R}^d, \\ \mathbf{z}_t &= [\tilde{\mathbf{p}}_t^{\mathrm{scr}}; \mathbf{f}_t] \in \mathbb{R}^{d+7}, \\ \mathbf{Z}_t^{\mathrm{orig}} &= [\mathbf{z}_{t-T+1}, \ldots, \mathbf{z}_t] \in \mathbb{R}^{T \times (d+7)}, \end{aligned} \tag{2}$$

where $T$ is the temporal window length, $d$ is the compressed descriptor dimension, and $\tilde{\mathbf{p}}_t^{\mathrm{scr}}$ denotes the normalized 7-D pose vector. We also construct a companion stream $\mathbf{Z}_t^{\mathrm{pert}} = \mathrm{Shuffle}(\mathbf{Z}_t^{\mathrm{orig}})$ using a lightweight feature-channel shuffle inspired by ShuffleNet [27]. This perturbation stream preserves frame order while disturbing feature channels, providing a consistency probe for the fusion module.

The original and perturbed streams are encoded by shared Mamba blocks [11] and fused by cross-attention. Denoting this encoder by $\Phi_{\mathrm{MCA}}(\cdot,\cdot)$, the reliability score is:

$$\begin{aligned} \mathbf{u}_t &= \Phi_{\mathrm{MCA}}\left(\mathbf{Z}_t^{\mathrm{orig}}, \mathbf{Z}_t^{\mathrm{pert}}\right), \\ c_t &= \sigma(\mathrm{MLP}\left(\mathrm{Pool}(\mathbf{u}_t)\right)), \end{aligned} \tag{3}$$

where $\mathrm{Pool}(\cdot)$ denotes temporal average pooling, $\mathrm{MLP}(\cdot)$ is a two-layer perceptron, and $\sigma(\cdot)$ is the sigmoid function. The output $c_t$ is a soft pose-reliability score, not a hard known/unknown scene label. Implementation details are provided in Appendix B.2.

### 3.3 Confidence-Guided Localization Policy

Mode selection is formulated as a sequential decision problem because each action depends on recent confidence, available reference poses, and accumulated odometric drift. We use the state

$$\mathbf{s}_t = [c_t, c_{t-1}, \ldots, c_{t-T_h+1}, \boldsymbol{\psi}_t], \quad \boldsymbol{\psi}_t = [m_{1,t}, \delta_{1,t}, \ldots, m_{K,t}, \delta_{K,t}], \tag{4}$$

where $T_h$ is the history length, $m_{k,t}$ marks whether the $k$-th candidate reference pose is valid, and $\delta_{k,t}$ is its normalized span to the current frame. The action space is $\mathcal{A} = \{0, 1, \ldots, K\}$: $a_t = 0$ accepts $\mathbf{p}_t^{\mathrm{scr}}$, while $a_t = k$ composes the $k$-th reference pose with odometry as in Eq. (1). For the action-conditioned pose $\hat{\mathbf{p}}_t(a_t) = (\hat{\mathbf{t}}_t(a_t), \hat{\mathbf{q}}_t(a_t))$ and ground truth $\mathbf{p}_t^* = (\mathbf{t}_t^*, \mathbf{q}_t^*)$, the reward is

$$\mathcal{R}(\mathbf{s}_t, a_t) = -\left(w_T \ell_t^T + w_R \ell_t^R + w_d \ell_t^d\right), \tag{5}$$

where $\ell_t^T = \|\hat{\mathbf{t}}_t(a_t) - \mathbf{t}_t^*\|_2$, $\ell_t^R = 2\arccos(|\langle \hat{\mathbf{q}}_t(a_t), \mathbf{q}_t^* \rangle|)$, and $\ell_t^d$ penalizes odometric span: $\ell_t^d = 0$ for $a_t = 0$ and $\ell_t^d = t - \tau_k$ when $a_t = k > 0$. The scalar weights $w_T$, $w_R$, and $w_d$ balance immediate pose accuracy and accumulated drift; small-error cases are mapped to zero reward and large errors are clipped at $-5$. The policy is optimized with PPO [31] during training only; inference uses the learned policy without ground-truth poses or rewards.

### 3.4 Training Protocol

LINK is trained in stages. The SCR backbone is first trained on the known-scene subset of each target dataset. Following the LightLoc protocol [23], the SCR pose supervision loss is:

$$\mathcal{L}_{\mathrm{SCR}} = \frac{1}{M_{\mathrm{scr}}} \sum_{i=1}^{M_{\mathrm{scr}}} \|\mathbf{p}_i^{\mathrm{scr}} - \mathbf{p}_i^*\|_1, \tag{6}$$

where $M_{\mathrm{scr}}$ is the number of SCR training scans, $\mathbf{p}_i^{\mathrm{scr}}$ is the predicted 7-D pose, and $\mathbf{p}_i^*$ is the ground-truth pose. After the SCR backbone is frozen, the reliability estimator is trained with soft targets derived from frozen-SCR pose errors on reliability-training samples with ground-truth poses, so confidence measures localization reliability rather than scene membership. For each frame, we define the combined SCR pose error $e_t$ and map it to a soft confidence target using a Gaussian kernel:

$$e_t = \ell_t^{T,\mathrm{scr}} + \beta \ell_t^{R,\mathrm{scr}}, \qquad \hat{c}_t = \exp\left(-\frac{e_t^2}{2\sigma_e^2}\right), \tag{7}$$

where $\ell_t^{T,\mathrm{scr}}$ and $\ell_t^{R,\mathrm{scr}}$ are the translation and rotation errors of the SCR hypothesis, $\beta$ balances the two units, and $\sigma_e$ is estimated from empirical SCR error statistics on the reliability-training samples. The reliability estimator is supervised by mean-squared regression:

$$\mathcal{L}_{\mathrm{CE}} = \frac{1}{M_{\mathrm{ce}}} \sum_{t=1}^{M_{\mathrm{ce}}} (c_t - \hat{c}_t)^2, \tag{8}$$

where $M_{\mathrm{ce}}$ is the number of reliability-training samples. Finally, with the SCR backbone and reliability estimator fixed, the decision policy is optimized with PPO using the reward in Eq. (5). The clipped PPO objective, generalized advantage estimation, reference-pose state construction, and odometry and inference-time state details are provided in Appendix B.3 and Appendix B.4.

## 4 UrbanUnseen and Evaluation Protocols

**Evaluation labels and regimes.** We operationalize partial scene coverage at the route level. A test frame is labeled as known if its ground-truth position lies inside the spatial support used for training under the chosen protocol, and as unknown otherwise. These labels are used only for protocol construction and diagnostic reporting; they are not provided to LINK or to any baseline during inference. We evaluate three regimes. The fully known regime follows the conventional repeated-route setting, where the test route is sufficiently covered by training data. The controlled partially known regime is constructed on Oxford [3, 21] and NCLT [7] by retaining only selected spatial regions as known during training while evaluating on the full test trajectories. The UrbanUnseen regime provides route-level known/unknown transitions that occur naturally in the collected routes.

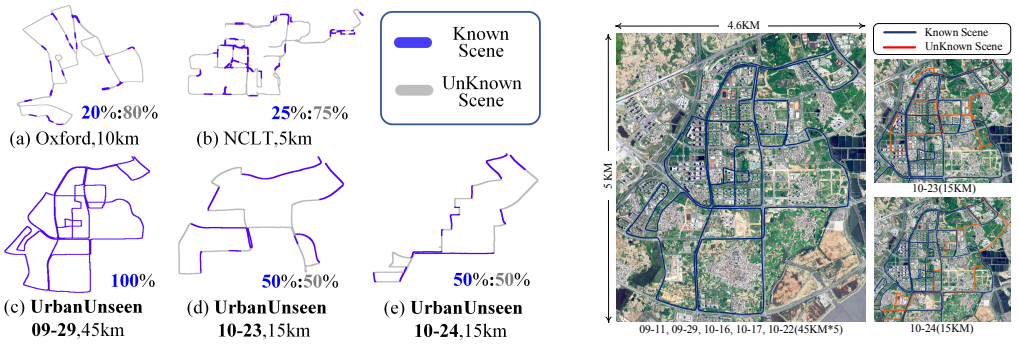



Figure 3: Dataset partition and UrbanUnseen benchmark visualization. Left: panels (a) and (b) show the controlled random partitions on Oxford and NCLT, while panels (c)–(e) show the route-level known/unknown distributions used in UrbanUnseen; blue denotes route segments retained as known scenes, and gray denotes unknown route segments outside the training support. Right: the UrbanUnseen benchmark overview on the satellite map, where the full-coverage routes are used for training and fully known evaluation, and routes 10-23 and 10-24 contain natural known-to-unknown scene transitions following the dark-blue/red legend.

**UrbanUnseen.** UrbanUnseen is a city-scale LiDAR relocalization benchmark designed to evaluate both repeated-route relocalization and route-level partial scene coverage. It covers approximately 23 $\mathrm{km}^2$ and contains seven trajectories. Five full-coverage routes, 09-11, 09-29, 10-16, 10-17, and 10-22, span approximately 44–47 km each. Two additional routes, 10-23 and 10-24, are 15 km long and each contains approximately 50% unknown scenes. We use 09-11, 10-16, 10-17, and 10-22 for training, 09-29 for fully known evaluation, and 10-23 and 10-24 for route-level partially known evaluation. The right side of Figure 3 provides a satellite-map overview of these routes and their known/unknown splits. All experiments in this paper use the Ouster OS1-128 LiDAR stream, with LiDAR-frame poses obtained by temporal alignment and interpolation from the 100 Hz FDI-Sigma navigation system. Additional acquisition details, sensor availability, trajectory statistics, and split definitions are provided in Appendix A.2.

**Controlled protocols on public benchmarks.** We evaluate controlled partial coverage on Oxford [3, 21] and NCLT [7]. For Oxford, all experiments use the trajectory-aligned ground-truth poses described in Appendix A.1. Since Oxford and NCLT do not explicitly separate training-covered and training-uncovered route segments, we construct two controlled partial-coverage protocols. In the random protocol, world-coordinate centers are sampled along the training trajectories, and scans within a 30 m radius are retained as known. In the manual protocol, known regions are selected around road intersections to approximate a deployment setting where data collection may concentrate around traffic-interaction areas rather than uniformly covering the route. In both protocols, the initial route segment is included in the known support to provide a valid global localization start. The average known-scene ratios are 20.0% on Oxford and 25.7% on NCLT under the random protocol, and 31.4% on Oxford and 22.1% on NCLT under the manual protocol. Panels (a)–(e) of Figure 3 visualize the main known/unknown partitions used in these protocols.

Exact public-benchmark sequence IDs, per-sequence known-scene ratios, and additional partition visualizations are provided in Appendices A.3 and A.4.

# 5 Experiments

We evaluate LINK under conventional fully known relocalization, controlled partial coverage on public benchmarks, and route-level partial coverage in UrbanUnseen. The analysis then examines the contribution of reliability-aware switching, temporal reliability estimation, reward design, and the chosen backbone/odometry components.

Table 1: Results on Oxford under fully known and random partial-coverage protocols.

| Protocol | Method | 15-13-06-37↓ | 17-13-26-39↓ | 17-14-03-00↓ | 18-14-14-42↓ | Average [m/°]↓ |
|---|---|---|---|---|---|---|
| **Fully Known** | SGLoc (CVPR'23) [21] | 1.79/1.67 | 1.81/1.76 | 1.33/1.59 | 1.19/1.39 | 1.53/1.60 |
| | DiffLoc (CVPR'24) [22] | 2.03/**1.04** | 1.78/**0.79** | 2.05/**0.83** | 1.56/**0.83** | 1.86/**0.87** |
| | LiSA (CVPR'24) [41] | 0.94/1.10 | 1.17/1.21 | 0.84/1.15 | 0.85/1.11 | 0.95/1.14 |
| | LightLoc (CVPR'25) [23] | 0.82/1.12 | 0.85/1.07 | 0.81/1.11 | 0.82/1.16 | 0.83/1.12 |
| | **LINK** | **0.75**/1.12 | **0.70**/1.05 | **0.70**/1.08 | **0.73**/1.14 | **0.72**/1.10 |
| **Only 20.0% Scenes Known** | SGLoc+FG (CVPR'23) [21] | 7.03/4.16 | 7.41/4.53 | 7.30/4.21 | 7.09/4.11 | 7.21/4.25 |
| | DiffLoc+FG (CVPR'24) [22] | 7.23/4.19 | 6.86/3.84 | 6.90/4.33 | 7.01/3.97 | 7.00/4.08 |
| | LiSA+FG (CVPR'24) [41] | 6.44/3.81 | 7.28/4.20 | 7.35/4.10 | 6.78/4.00 | 6.96/4.03 |
| | LightLoc+FG (CVPR'25) [23] | 7.12/3.72 | 7.75/4.09 | 5.66/4.04 | 7.26/4.18 | 6.94/4.00 |
| | **LINK** | **2.26/3.43** | **2.77/3.72** | **2.16/3.91** | **1.77/3.52** | **2.24/3.64** |

Table 2: Results on NCLT under fully known and random partial-coverage protocols.

| Protocol | Method | 2012-02-12↓ | 2012-02-19↓ | 2012-03-31↓ | 2012-05-26↓ | Average [m/°]↓ |
|---|---|---|---|---|---|---|
| **Fully Known** | SGLoc (CVPR'23) [21] | 1.20/3.08 | 1.20/3.05 | 1.12/3.28 | 3.81/4.74 | 1.83/3.54 |
| | DiffLoc (CVPR'24) [22] | 0.99/2.40 | 0.92/2.14 | 0.98/2.27 | **1.88**/2.43 | **1.19/2.31** |
| | LiSA (CVPR'24) [41] | 0.97/**2.23** | 0.91/**2.09** | 0.87/**2.21** | 3.30/2.84 | 1.51/2.34 |
| | LightLoc (CVPR'25) [23] | 0.98/2.76 | 0.89/2.51 | 0.86/2.67 | 3.86/3.51 | 1.64/2.86 |
| | **LINK** | **0.75**/2.31 | **0.66**/2.22 | **0.76**/2.57 | 2.94/**2.39** | 1.27/2.37 |
| **Only 25.7% Scenes Known** | SGLoc+FG (CVPR'23) [21] | 5.47/6.83 | 5.25/6.36 | 5.04/6.97 | 7.29/7.31 | 5.76/6.87 |
| | DiffLoc+FG (CVPR'24) [22] | 4.65/6.28 | 4.15/5.87 | 3.48/4.29 | 6.41/5.48 | 4.67/5.48 |
| | LiSA+FG (CVPR'24) [41] | 4.90/5.72 | 5.61/5.63 | 4.55/5.60 | 6.20/5.31 | 5.32/5.57 |
| | LightLoc+FG (CVPR'25) [23] | 3.61/5.84 | 3.62/3.82 | 3.55/4.38 | 6.36/5.69 | 4.29/4.93 |
| | **LINK** | **2.27/2.85** | **2.22/2.84** | **2.42/3.27** | **2.66/3.17** | **2.39/3.03** |

### 5.1 Experimental Setup

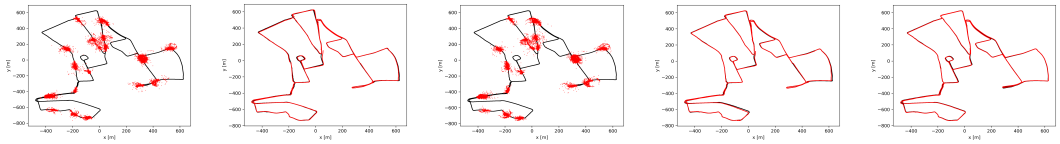
**Datasets, protocols, and metrics.** Following Section 4, we evaluate LINK on Oxford [3, 21], NCLT [7], and UrbanUnseen under fully known, controlled partial-coverage, and route-level partial-coverage regimes. Known/unknown labels are used only for protocol construction and diagnostics, and are not provided to any method during inference. Unless otherwise stated, results are reported as translation / rotation error [m/°], where lower is better; **bold** and underlining denote the best and second-best values in each comparison column.

**Baselines.** We compare with representative map-free LiDAR relocalization methods: SGLoc [21], DiffLoc [22], LiSA [41], and LightLoc [23]. In partially known settings, we also evaluate stronger "+FG" variants, where each baseline is combined with the same voxelized GICP odometry [19] and factor-graph optimization [37].

**Implementation details.** Following the LightLoc setup, the scene-agnostic SCR encoder is frozen and only the regression head is retrained on the known-scene subset using AdamW with a one-cycle learning-rate schedule from $5 \times 10^{-4}$ to $5 \times 10^{-3}$ for 50 epochs and a batch size of 256. The SCR voxel size is $0.25$ for Oxford, $0.3$ for NCLT, and $0.5$ for UrbanUnseen; odometry is estimated with voxelized GICP using a voxel size of $0.7$. The reliability estimator uses a temporal window of $T = 10$ and three shared Mamba blocks with widths $\{128, 64, 32\}$. The policy uses ten actions, including direct SCR localization and propagation from candidate reference poses, and is optimized with PPO. All experiments are conducted on two NVIDIA RTX 4090 GPUs. Additional details and the PPO objective are provided in Appendix B.

### 5.2 Main Results and Analysis

**Results on Oxford.** Oxford provides a controlled setting in which most test frames lie outside the retained training support. In the fully known protocol, LINK remains comparable to the leading relocalization baselines. Under 20.0% known-scene coverage, the lowest-error +FG baseline averages 6.94 m / 4.00°, whereas LINK averages 2.24 m / 3.64° in Table 1. The trajectory in Figure 4 helps explain this gap: factor-graph optimization reduces local discontinuities, but it cannot recover global consistency once the underlying SCR hypotheses move outside the covered scene distribution. Under



(a) LiSA (177 m, 42.3°) (b) LiSA† (7.2 m, 4.2°) (c) LightLoc (175 m, 49.5°) (d) LightLoc† (7.7 m, 4.0°) (e) **LINK** (2.7 m, 3.7°)

Figure 4: Oxford trajectory comparison on 17-13-26-39 under random partial coverage. † denotes factor-graph optimization; black and red lines denote ground truth and predictions.

Table 3: Results on UrbanUnseen under fully known and route-level partial-coverage protocols.

| **Method** | **Fully Known** | **Approx. 50% Scenes Known** | |
|---|---|---|---|
| | **09-29 [m/°]↓** | **10-23 [m/°]↓** | **10-24 [m/°]↓** |
| SGLoc+FG (CVPR'23) [21] | 8.15/5.72 | 14.53/9.42 | 14.50/8.36 |
| DiffLoc+FG (CVPR'24) [22] | 7.76/5.51 | 12.13/7.69 | 12.61/7.77 |
| LiSA+FG (CVPR'24) [41] | 7.38/5.29 | 13.97/7.32 | 12.88/7.98 |
| LightLoc+FG (CVPR'25) [23] | 5.85/4.64 | 10.43/6.94 | 9.58/7.17 |
| **LINK** | **5.47/4.23** | **7.24/6.15** | **7.52/6.42** |

Table 4: Policy ablation on Oxford under random partial coverage.

| **Policy** | **Acc. [%]↑** | **Recall [%]↑** | **F1 [%]↑** | **Average [m]↓** |
|---|---|---|---|---|
| Threshold-based [49] | 39.5 | 22.6 | 36.9 | 6.72 |
| Rule-based [26] | 62.5 | 52.0 | 68.4 | 4.38 |
| MLP-based [9] | 72.6 | 84.4 | 83.1 | 3.46 |
| **RL-based** | **94.3** | **98.8** | **96.3** | **2.24** |

Table 5: Temporal pose-reliability estimator ablation on NCLT under random partial coverage.

| **Variant** | **Temp.** | **Mamba** | **Cross-Attn.** | **Error [m/°]↓** |
|---|---|---|---|---|
| Temporal only | ✓ | | | 5.56/5.29 |
| + Mamba | ✓ | ✓ | | 4.28/4.37 |
| + Cross-attn. | ✓ | | ✓ | 4.40/4.33 |
| **Full** | ✓ | ✓ | ✓ | **2.39/3.03** |

the manual protocol, where known regions are concentrated around intersections, LINK averages 1.37 m / 2.48°; full per-sequence results are provided in Appendix C.1.

**Results on NCLT.** NCLT tests the same controlled partial-coverage question on a campus-scale route with different geometry and data collection conditions. The fully known results remain competitive, with LINK giving the best translation error on three of the four test sequences in Table 2. Under 25.7% known-scene coverage, the average error decreases from 4.29 m / 4.93° for LightLoc+FG to 2.39 m / 3.03° for LINK. This consistency with Oxford supports the same interpretation: once direct SCR predictions leave the covered scene distribution, post-hoc smoothing is less effective than selecting when to use regression and when to propagate from reliable references. Under the manual protocol, LINK averages 3.19 m / 3.63° on NCLT, with full per-sequence results in Appendix C.2.

**Results on UrbanUnseen.** Unlike the controlled Oxford and NCLT protocols, UrbanUnseen evaluates natural route-level known-to-unknown transitions. Absolute errors are larger than on Oxford and NCLT, but the same failure mode appears: stabilized SCR baselines still drift when the route contains genuinely unseen segments, whereas LINK improves over LightLoc+FG on both partially known routes and also gives the lowest error on the fully known 09-29 route in Table 3. The remaining errors on 10-23 and 10-24 make clear that route-level partial-coverage relocalization is harder than the controlled protocols, but the relative gap suggests that reliability-guided switching is not merely exploiting a favorable random repartition. Full trajectory-level diagnostics on UrbanUnseen are provided in Appendix D.1, and additional confidence visualizations are provided in Appendix D.2.

### 5.3 Ablation and Deployment Analysis

**Ablation of reliability-aware decision policy.** To evaluate whether LINK benefits from sequential mode selection rather than frame-wise confidence rejection, we compare the RL policy with threshold-based [49], rule-based [26], and MLP-based [9] alternatives. As shown in Table 4, the RL policy achieves the highest oracle-action agreement and reduces the Oxford translation error from 3.46 m for the strongest non-RL policy to 2.24 m. This indicates that using temporal context and reference-pose availability is more effective than treating confidence as an isolated per-frame signal.

**Ablation of temporal reliability estimator.** To examine what information is needed for robust switching, we ablate the temporal pose-reliability estimator. As shown in Table 5, the full estimator

Table 6: Policy reward ablation on Oxford under random partial coverage.

| Reward | Trans. | Rot. | Penalty | Error [m/°]↓ |
|---|---|---|---|---|
| Translation only | ✓ | | | 3.11/4.06 |
| + Rotation | ✓ | ✓ | | 2.71/3.85 |
| + Odometry penalty | ✓ | | ✓ | 2.55/3.80 |
| **Full reward** | ✓ | ✓ | ✓ | **2.24/3.64** |

Table 7: Odometry ablation under partial scene coverage. We report the average error [m/°] ↓.

| Method | Dataset | | | Real time |
|---|---|---|---|---|
| | Oxford↓ | NCLT↓ | UrbanUnseen↓ | |
| LINK(Pin-SLAM [28]) | 2.88/4.26 | 3.71/4.50 | 10.69/7.35 | × |
| LINK(Fast-LIO [39]) | 2.56/3.88 | 3.24/3.97 | 9.35/6.85 | ✓ |
| **LINK(GICP [19])** | **2.24/3.64** | **2.39/3.03** | **7.38/6.28** | ✓ |

Table 8: Backbone ablation under fully known and partially known coverage. Cells report average translation / rotation error [m/°]; lower is better.

| Dataset | Protocol | LINK w/ SGLoc [21] | LINK w/ DiffLoc [22] | LINK w/ LiSA [41] | LINK w/ LightLoc [23] |
|---|---|---|---|---|---|
| **Oxford** | **Fully Known** | 1.31/1.38 | 1.65/**0.83** | 0.88/1.03 | **0.72**/1.10 |
| | **20.0% Known** | 2.70/3.89 | 2.55/3.70 | 2.37/3.81 | **2.24/3.64** |
| **NCLT** | **Fully Known** | 1.62/3.29 | **1.14/2.10** | 1.43/2.31 | 1.25/2.37 |
| | **25.7% Known** | 4.20/5.55 | 3.92/4.68 | 3.97/4.30 | **2.39/3.03** |
| **UrbanUnseen** | **Fully Known** | 7.48/5.37 | 6.55/4.58 | 6.72/4.66 | **5.47/4.23** |
| | **Approx. 50% Known** | 13.62/8.60 | 12.13/7.54 | 11.39/7.77 | **7.38/6.28** |

Table 9: Inference runtime and model-size comparison on Oxford and NCLT.

| Method | Reliability [ms]↓ | Policy [ms]↓ | Runtime [ms]↓ | | GPU↓ | Model↓ |
|---|---|---|---|---|---|---|
| | | | Oxford | NCLT | | |
| SGLoc+FG [21] | – | – | 38 | 75 | 2.4GB | 116MB |
| LiSA+FG [41] | – | – | 38 | 75 | 2.4GB | 116MB |
| LightLoc+FG [23] | – | – | 29 | 45 | 2.1GB | 70MB |
| **LINK** | 8 | 11 | 48 | 64 | 2.5GB | 86MB |

achieves the lowest NCLT error when temporal modeling and pose–feature cross-attention are used together. This result suggests that reliability estimation benefits from modeling how pose hypotheses and scene features evolve jointly over time.

**Ablation of policy reward design.** We further ablate the policy reward to test whether the policy should optimize long-horizon pose consistency rather than instantaneous translation alone. As shown in Table 6, adding rotation and odometric-span penalties to translation supervision reduces the Oxford error from 3.11 m / 4.06° to 2.24 m / 3.64°. These terms help penalize decisions that are locally plausible but drift-prone under long odometric propagation.

**Ablation of odometry source.** To separate the switching mechanism from the relative-motion source, we evaluate different odometry modules. As shown in Table 7, GICP provides the best accuracy–runtime trade-off among Pin-SLAM [28], Fast-LIO [39], and voxelized GICP [19] across Oxford, NCLT, and UrbanUnseen. We therefore use GICP as the default odometry source in LINK.

**Ablation of SCR backbone.** We also evaluate whether the mode-selection framework depends on a particular SCR backbone. As shown in Table 8, LINK transfers across SGLoc, DiffLoc, LiSA, and LightLoc backbones, while LightLoc gives the strongest partial-coverage results among the tested variants. We therefore use LightLoc as the default SCR backbone.

**Deployment cost.** Finally, we compare deployment cost using runtime and model-size statistics on Oxford and NCLT. As shown in Table 9, the temporal pose-reliability estimator and learned decision policy add 8 ms and 11 ms, respectively, while LINK preserves real-time inference on both datasets. Since LINK does not introduce explicit map storage, its deployment cost mainly comes from learned reliability estimation and policy inference rather than a return to map-based retrieval.

# 6 Conclusion

We study map-free LiDAR relocalization under partial scene coverage, where test routes may deviate from training data. We propose LINK, a confidence-guided hybrid framework that evaluates pose reliability from SCR sequences to adaptively switch between direct localization and odometric propagation. To benchmark this, we introduce UrbanUnseen and controlled protocols on Oxford/NCLT datasets. Results demonstrate that LINK significantly enhances robustness in unseen areas while maintaining competitive accuracy in known scenes, highlighting the value of reliability-aware mode selection for practical deployment.

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# Appendix

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# A Benchmark and Protocol Details

## A.1 Oxford Ground-Truth Pose Alignment

For Oxford [3], we follow SGLoc [21] and use trajectory-aligned ground-truth poses for evaluation. The original repeated-route poses can contain noticeable discrepancies at the same road locations, which can confound frame-level localization errors when repeated traversals are compared directly. The aligned poses reduce this discrepancy while preserving the standard Oxford sequence split used by prior map-free LiDAR relocalization work. All Oxford experiments in this paper, including baselines and LINK, are evaluated under the same aligned poses.

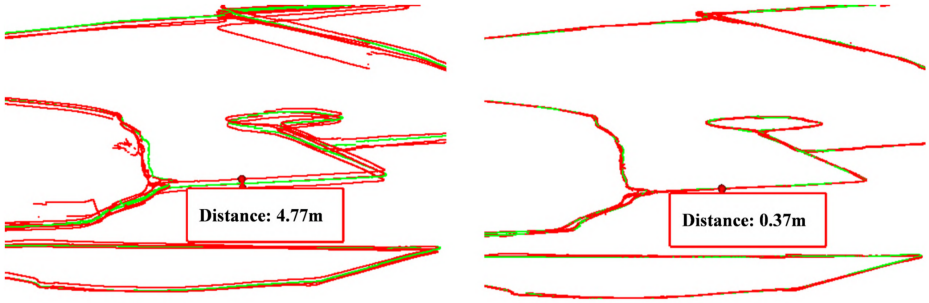



Figure 5: Oxford ground-truth pose alignment. The left panel shows repeated-route pose discrepancy in the original Oxford poses, while the right panel shows the trajectory-aligned poses used in our Oxford experiments. Green denotes the reference trajectory, and red denotes the corresponding trajectory before or after alignment. In this example, the local discrepancy decreases from 4.77 m to 0.37 m.

## A.2 UrbanUnseen Acquisition and Splits

Figure 6 summarizes the UrbanUnseen acquisition platform and route coverage. The dataset was collected with a multi-sensor vehicle equipped with an Ouster OS1-128 LiDAR, HIKROBOT cameras, Robosense RubyPlus-128 and M1-64 LiDARs, and an FDI-Sigma navigation system. The route map illustrates the full-coverage trajectories used for training and fully known evaluation, together with the two partial-coverage trajectories used to evaluate route-level known/unknown transitions.

Table 10 compares UrbanUnseen with representative outdoor LiDAR localization datasets. The native unknown-route evaluation column refers to whether the dataset protocol includes real test-route segments outside the training-covered route support, rather than whether such segments can be simulated by repartitioning an existing repeated-route benchmark.

Table 10: Comparison with representative outdoor LiDAR localization datasets. Route length and coverage area follow the representative benchmark-scale statistics reported by the corresponding dataset papers. Native unknown-route evaluation indicates whether the dataset includes real test-route segments absent from the training-covered route support in its intended evaluation protocol.

| Dataset | Rep. Route Length | Coverage Area | Native Unknown-Route Eval. | Repeated Trajectories | RGB Camera |
|---|---|---|---|---|---|
| KITTI [10] | 4 km | – | × | × | ✓ |
| Boreas [6] | 8 km | 6 $km^2$ | × | ✓ | ✓ |
| BiLiLo [43] | 10 km | 3 $km^2$ | × | ✓ | × |
| NCLT [7] | 5.5 km | 0.5 $km^2$ | × | ✓ | ✓ |
| Oxford [3] | 10 km | 2.5 $km^2$ | × | ✓ | ✓ |
| **UrbanUnseen** | **45 km** | **23** $km^2$ | ✓ | ✓ | ✓ |

UrbanUnseen contains seven trajectories: 09-11, 09-29, 10-16, 10-17, 10-22, 10-23, and 10-24. Table 11 lists the trajectory length, average vehicle speed, Ouster point-cloud count, and sensor availability for each trajectory. The Robosense RubyPlus-128 and Robosense M1-64 streams are unavailable for 09-29 due to equipment failure. All experiments in this paper use the Ouster OS1-128 LiDAR stream.

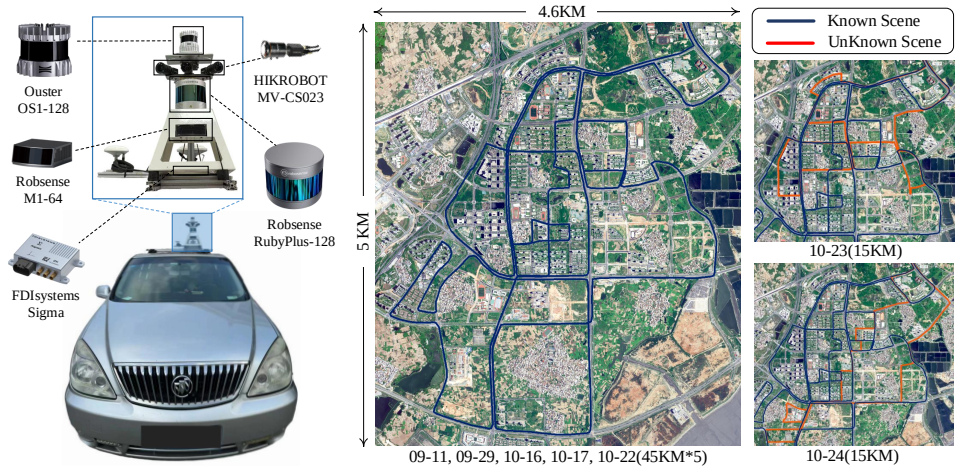



Figure 6: Overview of UrbanUnseen acquisition and route coverage. Left: collection vehicle and recorded sensor streams. Right: route coverage used in the UrbanUnseen split. The full-coverage routes are used for training and fully known evaluation, while trajectories 10-23 and 10-24 contain approximately 50% route segments outside the training coverage. Blue and red denote known and unknown route segments under the UrbanUnseen split.

Table 11: UrbanUnseen trajectory and sensor parameters. We report the trajectory length, average vehicle speed, and the number of point clouds collected by the Ouster OS1-128 stream. Sensor availability is indicated for each trajectory.

| Trajectory | Ouster OS1-128 | | | HIKROBOT camera (10 Hz) | Robosense RubyPlus-128 (10 Hz) | Robosense M1-64 (10 Hz) | FDI-Sigma (100 Hz) |
|---|---|---|---|---|---|---|---|
| | Length (km) | Speed (km/h) | Point clouds (K) | | | | |
| 09-11 | 45 | 26 | 62 | ✓ | ✓ | ✓ | ✓ |
| 09-29 | 47 | 27 | 62 | ✓ | × | × | ✓ |
| 10-16 | 44 | 25 | 64 | ✓ | ✓ | ✓ | ✓ |
| 10-17 | 47 | 26 | 65 | ✓ | ✓ | ✓ | ✓ |
| 10-22 | 46 | 26 | 63 | ✓ | ✓ | ✓ | ✓ |
| 10-23 | 15 | 34 | 15 | ✓ | ✓ | ✓ | ✓ |
| 10-24 | 15 | 35 | 15 | ✓ | ✓ | ✓ | ✓ |

FDI-Sigma provides ground-truth poses at 100 Hz, while the LiDAR streams run at 10–20 Hz. We therefore obtain the ground-truth pose for each LiDAR frame by temporally aligning the LiDAR timestamp to the FDI-Sigma trajectory and interpolating between neighboring navigation poses.

The UrbanUnseen split follows the route-level protocol used in the main paper. Trajectories 09-11, 10-16, 10-17, and 10-22 are used for training. Trajectory 09-29 is used for fully known evaluation. Trajectories 10-23 and 10-24 are used for route-level partially known evaluation; each contains approximately 50% unknown scenes under this split.

### A.3 Controlled Partial-Coverage Protocols on Oxford and NCLT

We use Oxford and NCLT to construct controlled partial-coverage protocols because they do not natively distinguish route segments covered during training from route segments absent from training. Known scenes and unknown scenes are used only as protocol labels for constructing the training subset and diagnostic reporting. These labels are not provided to LINK or to any baseline during inference.

For Oxford, the training sequences are 11-14-02-26, 14-12-05-52, 14-14-48-55, and 18-15-20-12, and the test sequences are 15-13-06-37, 17-13-26-39, 17-14-03-00, and 18-14-14-42. For NCLT, the training sequences are 2012-01-22, 2012-02-02, 2012-02-18, and 2012-05-11, and the test sequences are 2012-02-12, 2012-02-19, 2012-03-31, and 2012-05-26.

We use two controlled partition protocols. In the random protocol, world-coordinate centers are sampled on the ground-truth training trajectory, and scans within a 30 m radius of each center are retained as known scenes. In the manual protocol, road intersections are retained as known regions.

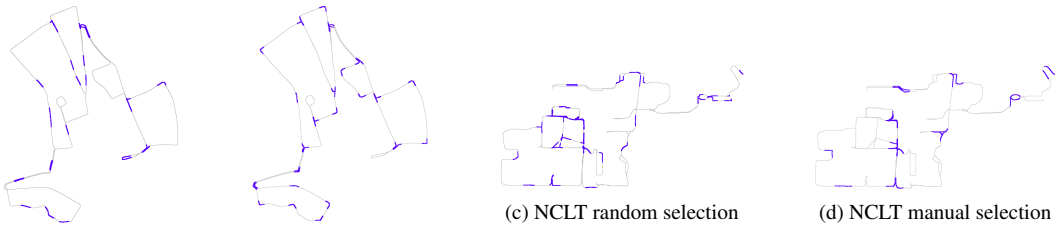
In both protocols, the initial route segment is treated as known by default to provide a valid globally referenced starting segment. Table 12 reports the sequence-level and average known-scene ratios.

Table 12: Controlled partition ratios for Oxford and NCLT. The table reports the proportion of training scans retained as known scenes under the random and manual protocols.

| Method | dataset | | Data | | | | |
|---|---|---|---|---|---|---|---|
| Random Selection | Oxford | proportion of Known Scenes | **11-14-02-26** | **14-12-05-52** | **14-14-48-55** | **18-15-20-12** | **Average** |
| | | | 20.9% | 20.4% | 21.5% | 20.5% | 20.0% |
| | NCLT | | **2012-01-22** | **2012-02-02** | **2012-02-18** | **2012-05-11** | **Average** |
| | | | 23.9% | 30.7% | 24.0% | 24.2% | 25.7% |
| Manual Selection | Oxford | | **11-14-02-26** | **14-12-05-52** | **14-14-48-55** | **18-15-20-12** | **Average** |
| | | | 36.7% | 30.0% | 34.2% | 25.0% | 31.4% |
| | NCLT | | **2012-01-22** | **2012-02-02** | **2012-02-18** | **2012-05-11** | **Average** |
| | | | 23.6% | 22.2% | 22.6% | 20.0% | 22.1% |

### A.4 Partition Visualizations

Figure 7 visualizes the known/unknown partitions for the two controlled protocols on Oxford and NCLT. In the random protocol, 25 centers are sampled for Oxford and 28 centers are sampled for NCLT. Because neighboring 30 m regions may overlap, the numbers of visibly separated known regions are 18 and 22, respectively. In the manual protocol, the retained known regions correspond to 23 key intersections on Oxford and 10 key intersections on NCLT.



(a) Oxford random selection

(b) Oxford manual selection

(c) NCLT random selection

(d) NCLT manual selection

Figure 7: Controlled partition visualization. The figure shows known/unknown partitions for Oxford and NCLT under the controlled partial-coverage protocols. Known scenes are shown in blue and unknown scenes in gray.

## B Implementation Details

### B.1 SCR Backbone and Training

LINK instantiates its SCR backbone with LightLoc [23]. The scene-agnostic encoder is kept frozen, and only the regression head is retrained on the known-scene subset of each target dataset. Under the partial-coverage protocols, the retraining set contains only the route segments designated as known by the corresponding protocol; segments designated as unknown are excluded from SCR retraining.

Training proceeds in three stages. The SCR regression head is trained first using Eq. (6). The temporal reliability estimator is then trained using pose hypotheses and intermediate features from the frozen SCR backbone. Finally, the localization policy is trained with the SCR backbone and reliability estimator fixed.

For the SCR regression head, the point cloud is encoded into a 512-dimensional scene descriptor following the LightLoc implementation. The descriptor is normalized to the unit sphere, augmented with Gaussian noise, normalized again, and concatenated before regression. The regression head is trained with a batch size of 256. The voxel size is set to $0.25$ for Oxford, $0.3$ for NCLT, and $0.5$ for UrbanUnseen. The intermediate feature map produced by the frozen encoder is retained for temporal reliability estimation.

### B.2 Temporal Reliability Estimator

The forward computation of the temporal reliability estimator follows Eqs. (2) and (3). We use a temporal window of $T = 10$ frames. Each temporal token concatenates the normalized 7-D SCR pose with the compressed feature map produced by $G(\cdot)$. The original temporal stream is paired with a shuffled perturbation stream that preserves frame order while perturbing the pose–feature tokens. The two streams are encoded by shared Mamba blocks [11], fused by cross-attention, temporally averaged, and mapped by a two-layer MLP with sigmoid activation.

Training uses the soft target and regression objective in Eqs. (7) and (8). The reliability estimator is trained with Adam [18] for 50 epochs using a batch size of 64, with the learning rate decayed from $1 \times 10^{-4}$ to $1 \times 10^{-6}$. The resulting score is used as the pose-reliability signal for policy selection.

### B.3 Localization Policy and Training Objectives

The main text defines the policy state, action space, action-conditioned pose rule, and reward in Eqs. (4), (1), and (5). This subsection records the implementation-level state representation and the PPO objective used to train the policy.

The reference-pose part of the policy state is a fixed-size candidate summary, $\boldsymbol{\psi}_t = [m_{1,t}, \delta_{1,t}, \ldots, m_{K,t}, \delta_{K,t}]$. For each candidate, $m_{k,t}$ indicates whether a valid historical reference pose is available, and $\delta_{k,t}$ is the normalized temporal span from that reference pose to the current frame. This summary is concatenated with the recent confidence history defined in Eq. (4). In the reported setting, the policy has ten actions: one direct-SCR action and nine reference-pose propagation actions. Ground-truth poses and rewards are used only during policy training; inference uses the learned policy and does not require ground-truth poses, rewards, or known/unknown labels.

For compact notation, let

$$\mathcal{R}_t = \mathcal{R}(\mathbf{s}_t, a_t), \tag{9}$$

where $\mathcal{R}(\mathbf{s}_t, a_t)$ follows the reward definition in Eq. (5). PPO [31] is used to train the policy for 200 episodes by maximizing the expected cumulative reward:

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t \mathcal{R}_t\right], \tag{10}$$

where $\pi_\theta(a_t|\mathbf{s}_t)$ is the policy parameterized by $\theta$ and $\gamma$ is the discount factor.

The temporal-difference residual $\Delta_t^V$ and advantage $\hat{A}_t$ are estimated with generalized advantage estimation:

$$\begin{aligned} \Delta_t^V &= \mathcal{R}_t + \gamma V_\phi(\mathbf{s}_{t+1}) - V_\phi(\mathbf{s}_t), \\ \hat{A}_t &= \sum_{l=0}^{\infty} (\gamma\lambda)^l \Delta_{t+l}^V, \end{aligned} \tag{11}$$

where $V_\phi$ is the value function parameterized by $\phi$ and $\lambda$ is the GAE parameter. With the likelihood ratio

$$\rho_t(\theta) = \frac{\pi_\theta(a_t|\mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(a_t|\mathbf{s}_t)}, \tag{12}$$

where $\theta_{\text{old}}$ denotes the fixed policy parameters before the PPO update. The clipped PPO loss minimized during training is

$$\mathcal{L}_{\text{PPO}}^{\text{clip}} = -\mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{13}$$

where $\epsilon$ is the clipping parameter and $\mathbb{E}_t$ denotes empirical averaging over training timesteps. The value-function loss and entropy term are

$$\mathcal{L}_V = \mathbb{E}_t\left[\left(V_\phi(\mathbf{s}_t) - V_t^{\text{target}}\right)^2\right], \qquad \mathcal{L}_H = \mathbb{E}_t\left[\mathcal{H}(\pi_\theta(\cdot|\mathbf{s}_t))\right]. \tag{14}$$

Here, $V_t^{\text{target}}$ is the value-regression target and $\mathcal{H}(\cdot)$ denotes policy entropy. The final decision-policy objective is

$$\mathcal{L}_{\text{DM}} = \mathcal{L}_{\text{PPO}}^{\text{clip}} + c_1\mathcal{L}_V - c_2\mathcal{L}_H, \tag{15}$$

where $c_1$ and $c_2$ weight value fitting and entropy regularization.

### B.4 Odometry and Inference-Time State

The relative-motion term $\Delta\mathbf{p}_{\tau_k \to t}^{\mathrm{odom}}$ in Eq. (1) is estimated using GICP odometry. In the partially known experiments, the "+FG" variants of SCR baselines use the same GICP odometry source and factor-graph optimization. This setting assesses whether augmenting baseline SCR outputs with factor-graph optimization is sufficient to explain the improvement, while LINK performs learned frame-wise mode selection between direct SCR localization and odometric propagation during inference.

At inference time, LINK is map-free in the sense used throughout the paper: it does not store, retrieve, or query an explicit pre-built 3-D map database or map-feature database. The SCR backbone, reliability estimator, and policy are learned model parameters. The reference-pose memory and odometry estimates are runtime state used for pose propagation, not an explicit map database. Known/unknown protocol labels are not provided to LINK or to any baseline during inference. This map-free setting should therefore not be interpreted as stateless or zero-memory.

## C Additional Quantitative Results

This appendix complements the main experiments with additional evidence that is too detailed for the main paper. The manual protocols stress sparse coverage around road intersections, while the runtime and storage comparison clarifies the deployment cost of adding reliability estimation and policy inference. All localization entries follow the metric convention in Section 5.1: translation / rotation error [m/°], lower is better.

### C.1 Manual Protocol on Oxford

The Oxford manual protocol concentrates known regions around road intersections rather than distributing them uniformly along the route. The per-sequence results in Table 13 test whether the improvements in the random partial-coverage protocol persist under this nonuniform split.

Table 13: Manual-selection results on Oxford. The protocol uses 31.4% known-scene coverage for training and evaluates on the full test trajectories.

| Protocol | Method | 15-13-06-37↓ | 17-13-26-39↓ | 17-14-03-00↓ | 18-14-14-42↓ | Average [m/°]↓ |
|---|---|---|---|---|---|---|
| **Only 31.4% Scenes Known** | SGLoc [21] | 75.48/34.03 | 78.03/31.67 | 74.42/32.34 | 78.11/33.99 | 76.51/33.01 |
| | SGLoc+FG [21] | 8.21/5.26 | 8.28/6.24 | 8.34/6.60 | 8.06/6.47 | 8.22/6.14 |
| | DiffLoc [22] | 60.23/27.02 | 59.13/26.74 | 58.23/27.05 | 61.91/28.70 | 59.88/27.38 |
| | DiffLoc+FG [22] | 7.21/4.90 | 6.13/4.00 | 7.21/5.13 | 6.86/5.40 | 6.85/4.86 |
| | LiSA [41] | 68.63/29.72 | 71.31/30.14 | 69.22/30.67 | 71.07/29.44 | 70.06/29.99 |
| | LiSA+FG [41] | 7.03/5.35 | 7.22/5.90 | 7.59/5.44 | 7.34/5.81 | 7.30/5.63 |
| | LightLoc [23] | 78.54/32.93 | 70.07/29.69 | 70.84/34.73 | 70.12/30.59 | 72.39/31.99 |
| | LightLoc+FG [23] | 7.15/4.42 | 7.22/3.91 | 6.65/4.44 | 5.67/4.38 | 6.67/4.29 |
| | **LINK** | **1.46/2.57** | **1.02/1.97** | **1.27/2.58** | **1.71/2.79** | **1.37/2.48** |

### C.2 Manual Protocol on NCLT

The NCLT manual protocol provides the same sparse-coverage test on a campus-scale route. Table 14 gives the per-sequence breakdown used to support the main-paper manual-protocol summary.

### C.3 Runtime and Storage Analysis

The main paper summarizes inference runtime and model size on Oxford and NCLT in Table 9, where Reliability and Policy denote the same LINK inference components reported here. Table 15 provides a complementary sequence-specific NCLT view that further separates model storage, explicit map storage, and localization error.

Table 14: Manual-selection results on NCLT. The protocol uses 22.1% known-scene coverage for training and evaluates on the full test trajectories.

| Protocol | Method | 2012-02-12↓ | 2012-02-19↓ | 2012-03-31↓ | 2012-05-26↓ | Average [m/°]↓ |
|---|---|---|---|---|---|---|
| **Only 22.1% Scenes Known** | SGLoc [21] | 121.37/55.11 | 118.09/55.24 | 125.55/54.39 | 122.30/57.78 | 121.83/55.63 |
| | SGLoc+FG [21] | 7.17/7.23 | 7.76/7.38 | 6.94/6.82 | 7.39/7.40 | 7.32/7.21 |
| | DiffLoc [22] | 118.50/54.88 | 124.47/53.23 | 128.34/57.04 | 124.12/57.83 | 123.86/55.75 |
| | DiffLoc+FG [22] | 6.95/7.32 | 7.45/6.96 | 7.68/6.29 | 7.79/7.08 | 7.47/6.91 |
| | LiSA [41] | 123.41/58.32 | 127.06/60.33 | 119.90/59.35 | 121.61/57.72 | 123.00/58.93 |
| | LiSA+FG [41] | 6.91/6.72 | 5.94/6.40 | 6.70/6.94 | 6.73/6.31 | 6.57/6.59 |
| | LightLoc [23] | 120.11/56.40 | 123.68/58.72 | 132.70/58.58 | 127.55/58.19 | 126.01/57.97 |
| | LightLoc+FG [23] | 6.88/6.74 | 4.48/5.32 | 5.00/4.10 | 6.41/9.34 | 5.69/6.38 |
| | **LINK** | **2.58/3.65** | **3.51/4.06** | **1.84/2.83** | **4.81/3.96** | **3.19/3.63** |

Table 15: Runtime and storage comparison. The table reports runtime, memory, storage, and localization error on NCLT sequence 2012-05-26 under the random partially known protocol. Storage is reported as model storage plus explicit map storage; LINK uses 86 MB model storage and 0 MB explicit map storage.

| Method | Reliability [ms]↓ | Policy [ms]↓ | GPU Mem.↓ | Model↓ | Map↓ | Runtime↓ | Error [m/°]↓ |
|---|---|---|---|---|---|---|---|
| LightLoc+FG [23] | – | – | 2.1GB | 70MB | **0MB** | **45**ms | 6.36/5.69 |
| BEVplace++ [25] | – | – | 1.9GB | 17MB | 841MB | 54ms | 11.83/4.64 |
| **LINK** | 8 | 11 | 2.5GB | 86MB | **0MB** | 64ms | **2.66/3.17** |

# D Qualitative Diagnostics

## D.1 Trajectory-Level Comparisons

These trajectory diagnostics test whether the quantitative improvements correspond to route-level consistency rather than isolated frame-wise improvements. Figure 8 compares Oxford trajectories under the manual protocol, where known scenes are concentrated around road intersections.

The Oxford trajectories diagnose route-level consistency under sparse manual coverage. Direct LightLoc produces visible off-route errors under the manual protocol. LightLoc+FG reduces some drift, but does not fully remove route-level errors. LINK keeps the predicted trajectory closer to the ground truth over the full route, indicating that the improvement is not only an average-error effect.

UrbanUnseen provides a complementary diagnostic under route-level known/unknown transitions. Figure 9 includes both direct SCR errors and +FG variants so that the effect of LINK can be compared against raw and post-processed baselines.

The UrbanUnseen trajectories show that LINK's behavior extends from controlled public-benchmark partitions to route-level known/unknown transitions. This visualization is diagnostic rather than a substitute for the quantitative UrbanUnseen table, and the larger deviations on 10-23 and 10-24 indicate that route-level partial-coverage relocalization remains more difficult than controlled public-benchmark partitions.

The route-level plots above summarize complete trajectories. Figure 10 provides a complementary local view on the fully known UrbanUnseen 09-29 route, where the selected segment is nominally covered by training data. The example illustrates that local SCR instability can still appear in covered routes, and that LINK can correct such local drift through reliability-guided switching. This figure should be interpreted as a qualitative diagnostic rather than a separate quantitative benchmark.

## D.2 Confidence Visualizations

The confidence diagnostics examine whether the reliability estimator suppresses unreliable absolute SCR hypotheses before policy selection. On NCLT, Figure 11 compares raw SCR hypotheses with the high-confidence predictions retained by LINK under the random partially known protocol.

On UrbanUnseen, Figure 12 applies the same diagnostic to route-level transitions.

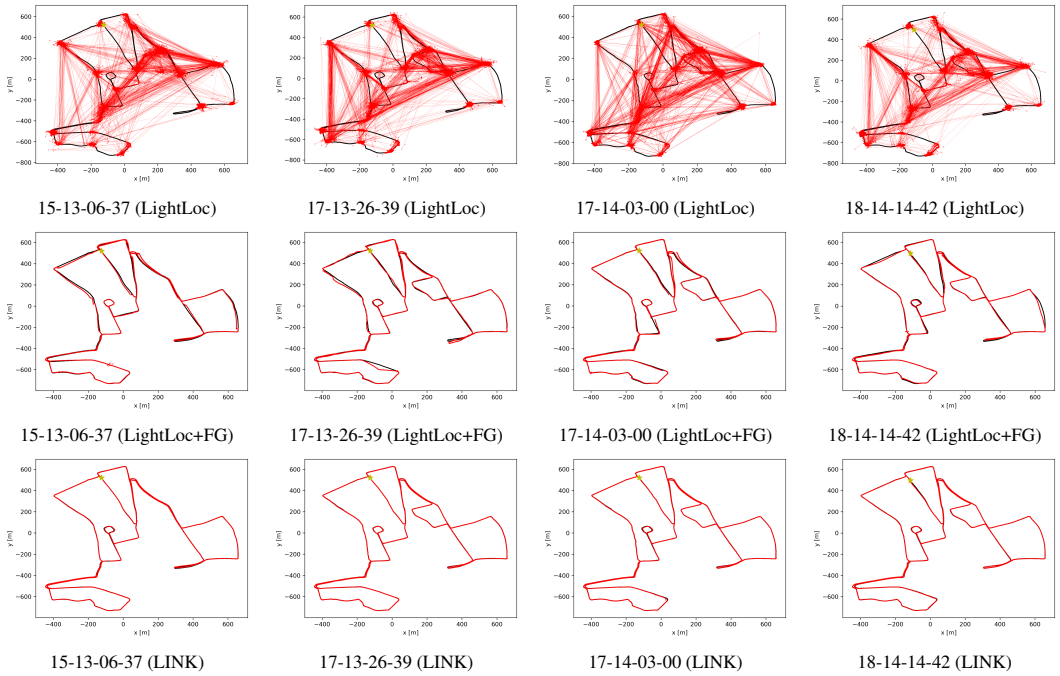



Figure 8: Oxford manual-coverage trajectory comparison. The figure shows trajectory-level results under the manual-selection protocol, where the known-scene coverage is 31.4%. Columns correspond to test sequences 15-13-06-37, 17-13-26-39, 17-14-03-00, and 18-14-14-42. Rows correspond to LightLoc, LightLoc+FG, and LINK, respectively. Black denotes the ground-truth trajectory, and red denotes the predicted trajectory.

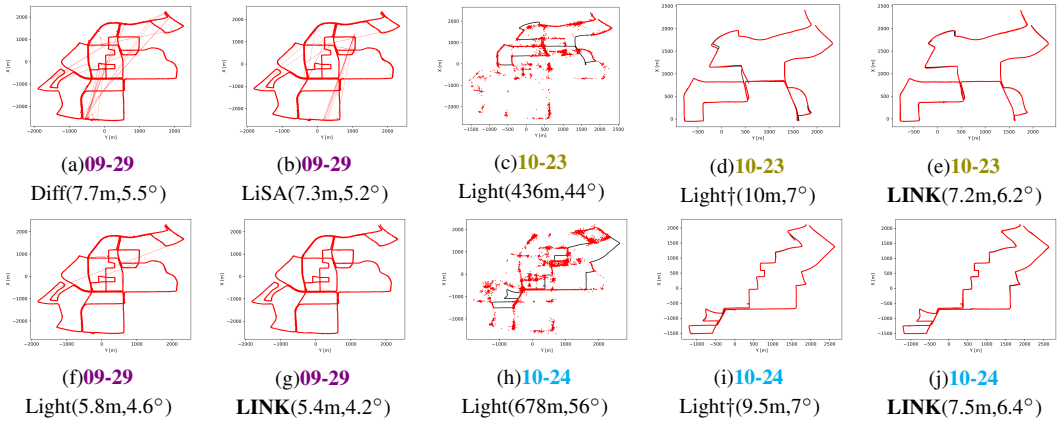
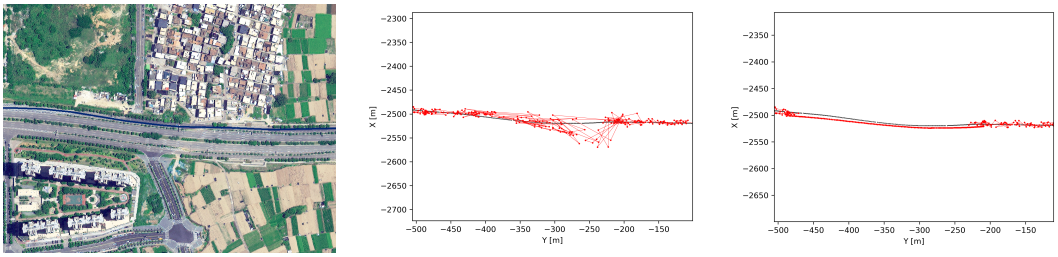



Figure 9: UrbanUnseen trajectory comparison. LINK is evaluated on trajectories **09-29** (fully known), **10-23** (approximately 50% scenes known), and **10-24** (approximately 50% scenes known). Ground truth and predictions are shown as black and red lines, respectively. † denotes factor-graph optimization.




(a) Remote-sensing view (b) 09-29 LightLoc (c) 09-29 LINK

Figure 10: UrbanUnseen local alignment detail on the fully known 09-29 route. Panel (a) shows the corresponding remote-sensing view, while panels (b) and (c) compare LightLoc and LINK on the selected road segment. This zoom-in complements the trajectory-level results by showing a local case where LightLoc exhibits drift and LINK remains better aligned.

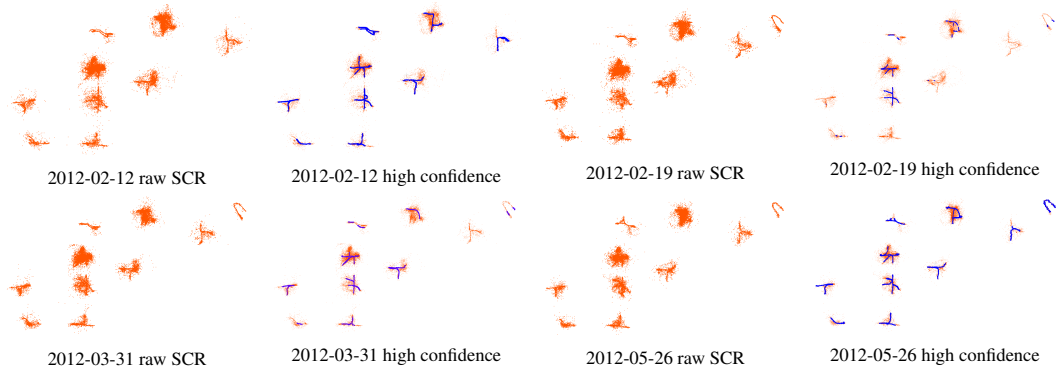



Figure 11: Confidence visualization on NCLT. Red denotes raw SCR hypotheses, and blue denotes high-confidence predictions retained by LINK. The figure diagnoses confidence behavior in complex scenes: confidence filtering suppresses many unreliable hypotheses, while some route segments may still receive low confidence because reliability is estimated as a soft signal rather than a perfect scene-membership label.

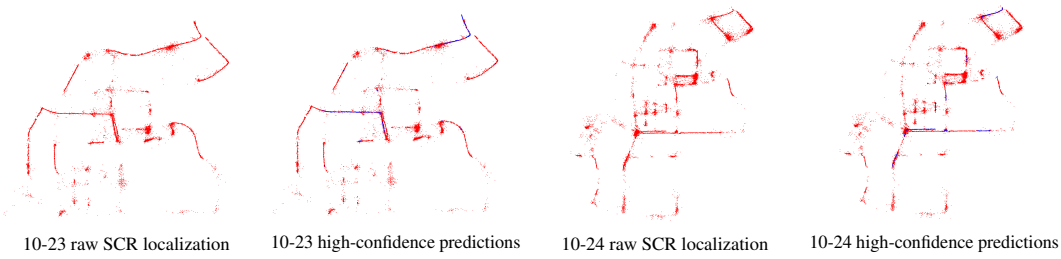



Figure 12: Confidence visualization on UrbanUnseen. Red denotes raw SCR hypotheses, and blue denotes high-confidence predictions retained by LINK. The diagnostic function of this figure is to show that raw SCR hypotheses can include numerically plausible but incorrect absolute predictions in unknown route segments, and confidence filtering suppresses many of these unreliable hypotheses before policy selection.

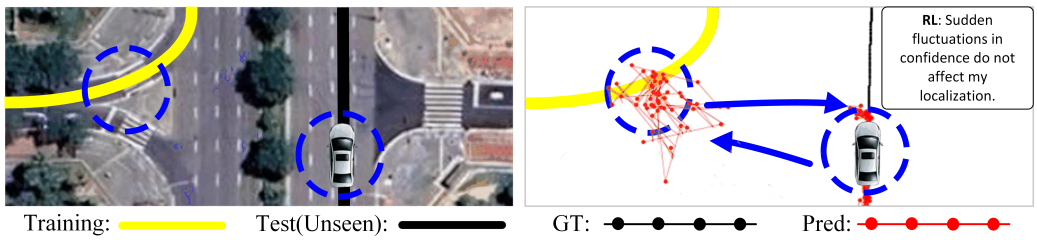



Figure 13: Ambiguous known/unknown passage. The yellow band denotes the training-covered route support, red points denote LINK predictions, and the black trajectory denotes the unseen test route. Brief passages near known support may produce confidence fluctuations, but the policy can still maintain a consistent globally referenced trajectory by selecting between direct SCR hypotheses and propagation from reliable historical poses.

The confidence maps should be interpreted as a soft pose-reliability signal, not a hard known/unknown classifier. In complex regions, some known segments may still receive low confidence, and brief passages near known route support may produce ambiguous confidence patterns, as illustrated in Figure 13. These ambiguities do not by themselves imply localization failure: confidence supports the decision policy by helping it avoid unreliable direct SCR predictions and by enabling propagation from a historical reference pose when the current absolute hypothesis is unreliable.

# E Limitations

LINK targets outdoor LiDAR relocalization under partial route coverage, and the claims in this paper are bounded by the Oxford, NCLT, and UrbanUnseen settings evaluated in Sections 4 and 5. Two operating assumptions matter in this setting: the system benefits from occasional reliable absolute SCR hypotheses to keep historical reference poses globally anchored, and relative odometry should not be asked to bridge arbitrarily long intervals without trustworthy absolute updates. The policy is designed around these assumptions by using confidence history, reference-pose availability, and an odometric-span penalty. The confidence score should also be read as a soft pose-reliability signal rather than a hard known/unknown scene label; ambiguous passages may produce confidence fluctuations, but these cases can still be handled by policy selection, as shown in Appendix D.2. These boundaries are consistent with the paper's scope, while broader sensor setups or deployment conditions beyond the evaluated outdoor LiDAR relocalization protocols require separate validation.

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